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Question 1: Tokenization

The word "MISUNDERSTANDING" is uncommon enough that a sub-word tokenizer would likely split it into smaller pieces rather than keep it whole.

(a) Split "MISUNDERSTANDING" into 3 reasonable sub-word pieces (there's no single "correct" answer — justify your split).

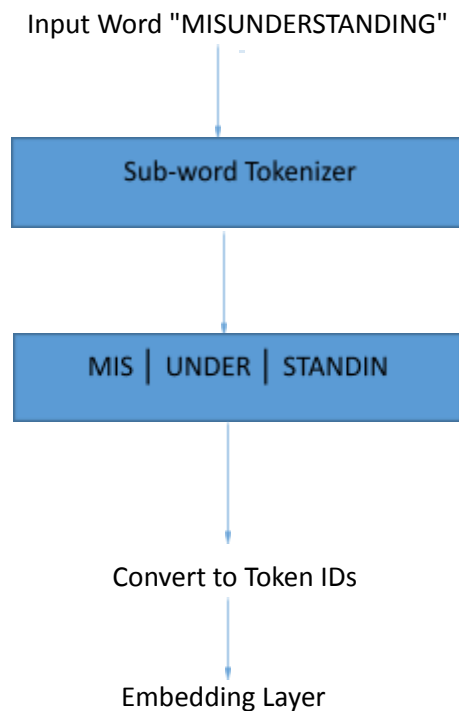
(b) In 1-2 sentences, explain WHY sub-word tokenization (rather than whole-word tokenization) helps a model handle a word like this one, even if the model has never seen the whole word before in training.

a) Split "MISUNDERSTANDING" into 3 sub-word pieces

Answer:

- MIS
- UNDER
- STANDING

b) Sub-word Tokenization Architecture



Question 2: Embeddings Consider the word pairs “happy” and “joyful” and “happy” and “chair”.

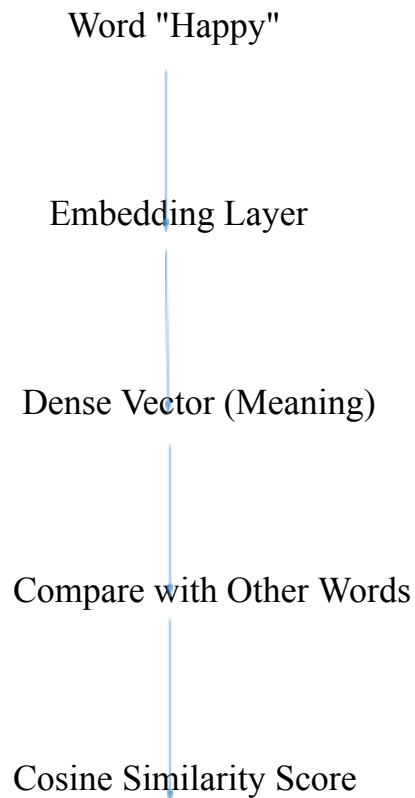
- (a) Which pair would you expect to have HIGHER cosine similarity in embedding space, and why?
- (b) In your own words, what does an embedding actually represent, and how is it different from a one-hot vector?

a) Which pair has higher cosine similarity?

Happy and Joyful have higher cosine similarity because they have similar meanings.

(b) What is an embedding? How is it different from a one-hot vector?

An embedding is a dense vector that represents the meaning of a word. A one-hot vector only identifies the word and does not store semantic relationships.



Question 3: Self-Attention Worked Example. Read the sentence: "Sam tried to call George on the phone, but he wasn't successful."

- (a) The word "he" should attend most strongly to which other word in the sentence? Circle one: "Sam" / "George"

(b) Explain in 2-3 sentences WHY self-attention (not a simple word-order rule) is needed to correctly resolve what "he" refers to here.

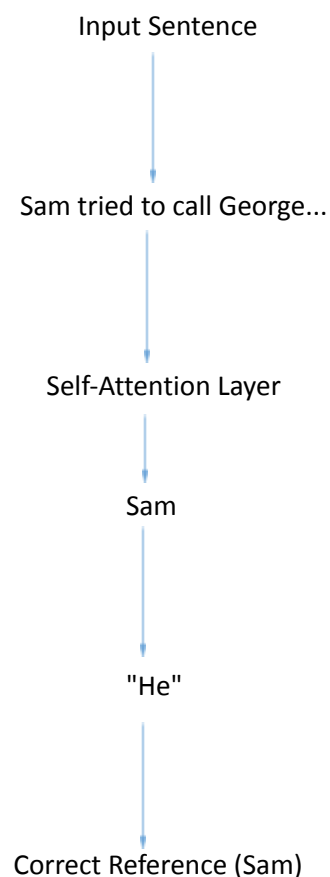
(a) "He" refers to:

✓ Sam

(b) Why is self-attention needed?

Self-attention looks at the relationships between all words in the sentence. It helps the model correctly identify that **"he" refers to Sam** based on the sentence's meaning, not just word order.

b) Self-Attention



Question 4: Query, Key, Value. In your own words, briefly explain what the Query, Key, and Value vectors each represent in self-attention, and how they combine to produce an attention score.

🎬 **Query (Q):** Finds the important words.

🎬 **Key (K):** Matches with the query.

🎬 **Value (V):** Provides the information.

Input Tokens



Linear Layers



Query Key Value



Compare



Attention Scores



Weighted Values



Final Output